

Socket Class Documentation

Source File: Socket.h
Namespace: cap
Class Header: `class Socket : public Object`

Overview

The *Socket* class stores the output of a *Dev* object at a selected bit of *Word* object.

Constructors

- `Socket()` (default constructor)
 - **Purpose:** Sets the word and circuit to null.
- `Socket(const Socket& obj)` (copy constructor) [deleted]

Destructor

- `~Socket()` [virtual]
 - **Purpose:** Removes the word and circuit.

Assignment Operator

- `operator=(const Socket& rhs)` [deleted]

Member Functions

- `destination(Word& obj)`
 - **Purpose:** Assigns *obj* as the word and makes the selected bit the most significant bit, and then returns true if no word was assigned; otherwise, it returns false.
 - **Parameters:**
 - *obj*: *Word* reference object.
 - **Return:** Boolean.
- `source(Dev& obj)`
 - **Purpose:** Assigns *obj* as the circuit, and then returns true if *obj* is valid and no circuit was assigned; otherwise, it returns false.
 - **Parameters:**
 - *obj*: *Dev* reference object.
 - **Return:** Boolean.
- `set(size_t idx)`
 - **Purpose:** Makes *idx* the selected bit of the word if word is not null and *idx* is less than the size of the word.
 - **Parameters:**
 - *idx*: Unsigned integer.
- `configured() const`
 - **Purpose:** Returns true if it has a word destination and a valid circuit source; otherwise, it returns false.
 - **Return:** Boolean.
- `unbind()`
 - **Purpose:** Assigns null to both the source and destination.
- `evaluate()`
 - **Purpose:** Assigns the output of the circuit to the selected bit of the word, and then returns true if it is configured; otherwise, it returns false.
 - **Return:** Boolean.
- `toString() const [final]`
 - **Purpose:** Returns "T" if the selected bit of its word is true, "F" if the selected bit of its word is false, or "X" if the word is not null.
 - **Return:** String.